

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE – CONTINGENCY**

3410UD0-1



Z22-3410UD0-1

THURSDAY, 23 JUNE 2022 – MORNING

CHEMISTRY – Unit 2:
Chemical Bonding, Application of Chemical Reactions
and Organic Chemistry

HIGHER TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
 You may use pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 4(c) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	11	
3.	13	
4.	11	
5.	11	
6.	7	
7.	10	
8.	8	
Total	80	

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Answer **all** questions.

1. (a) Crude oil can be separated into simpler mixtures called fractions. The table shows information about some of the main fractions obtained from crude oil by fractional distillation.

Fraction	Boiling point range (°C)	Number of carbon atoms present in the alkanes
petroleum gases	< 20	C ₁ –C ₄
petrol	30–75	C ₅ –C ₈
naphtha	70–180	C ₉ –C ₁₂
kerosene	180–250	C ₁₃ –C ₁₆
diesel oil	250–340	C ₁₇ –C ₂₀
lubricating oil	340–500	C ₂₀ –C ₂₄
fuel oil	490–580	C ₂₅ –C ₂₈

- (i) Decane has a boiling point of 174 °C. Give the name of the fraction which contains decane. [1]

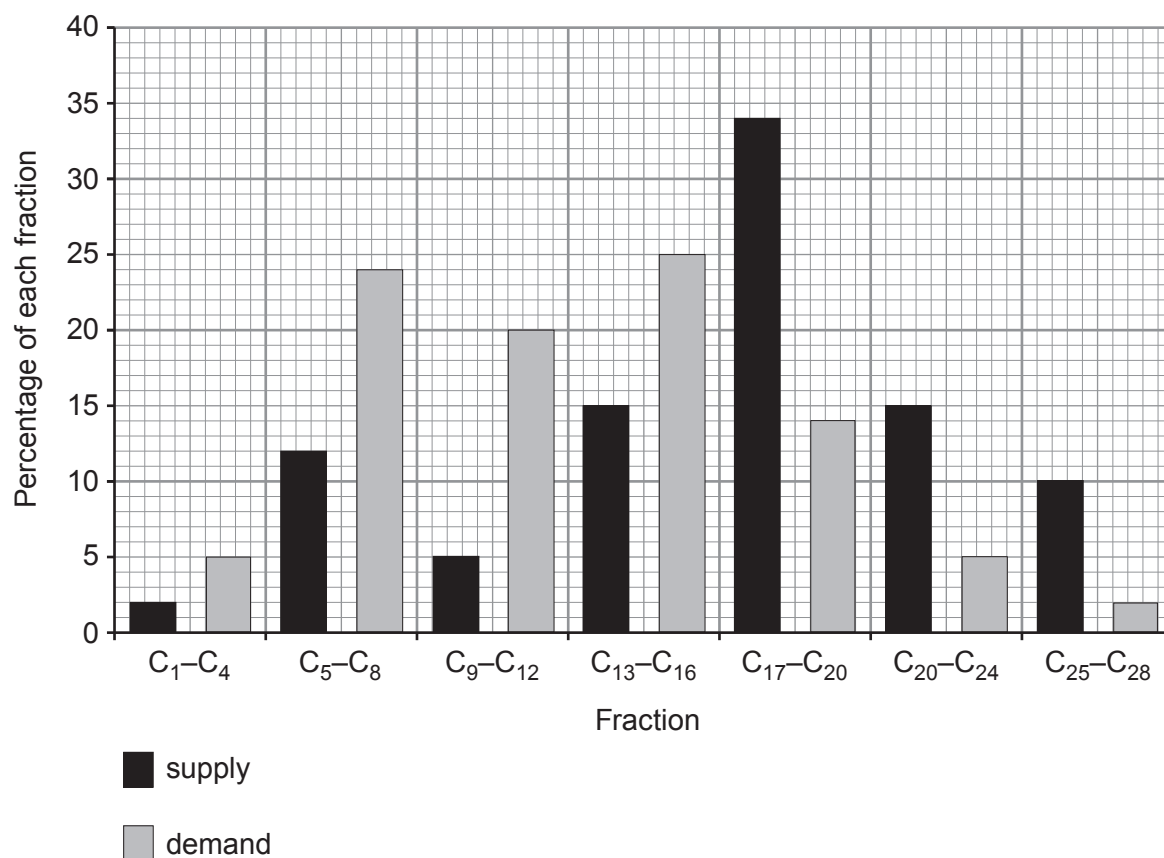
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- (ii) One alkane is found in both diesel oil and lubricating oil. Give the number of carbon atoms in this alkane. [1]

.....



- (b) The bar chart shows the supply and demand for some of the fractions obtained from crude oil.



- (i) Give the fraction where the demand is 100 % greater than the supply. [1]

.....

- (ii) Put a **tick** (✓) in the box next to the statement that best describes how the supply and demand of the fractions change as the chain length increases. [1]

supply is greater than demand for all fractions

☐

supply is greater than demand up to C₁₆ after which demand is greater than supply

☐

demand is greater than supply up to C₁₆ after which supply is greater than demand

☐

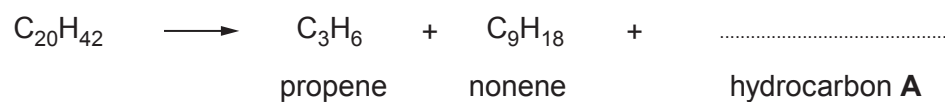
the difference between supply and demand increases up to C₁₆ after which it decreases

☐


- (iii) Oil companies have solved the problem of over-supply of some fractions using a process called cracking.

The alkane $C_{20}H_{42}$ can be cracked forming propene, nonene and **one molecule** of hydrocarbon **A**.

- I. Complete the equation for the cracking of $C_{20}H_{42}$. [1]



- II. Propene is an important raw material in the production of polypropene.

Give the name of the process used to make polypropene from propene. [1]

.....



- (c) Every year thousands of acres of forests are destroyed by wildfires. Fire-fighters use several different methods to put out this type of fire.



State **three** methods that are used to put out forest fires.

Give the part of the fire triangle being removed in each method.

Each method should refer to a different part of the fire triangle.

[3]

Method 1

Part of the fire triangle being removed

Method 2

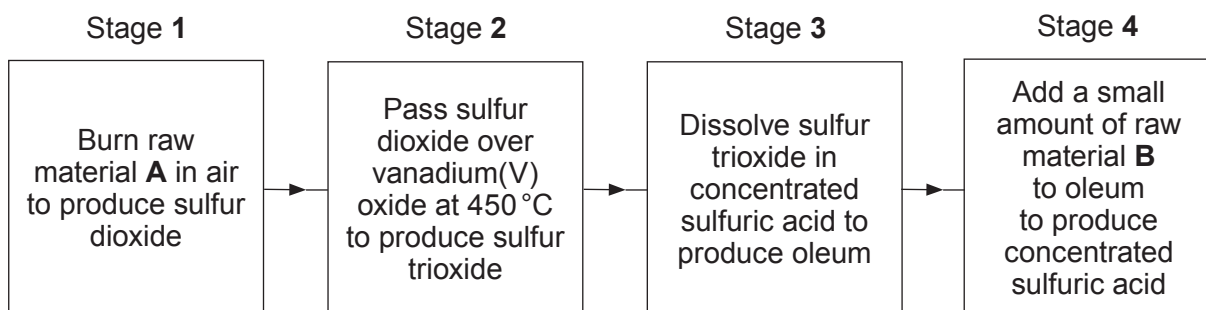
Part of the fire triangle being removed

Method 3

Part of the fire triangle being removed



2. (a) Sulfuric acid is produced by the contact process. The flow diagram shows the main stages in the process.



- (i) Give the name of raw materials **A** and **B**. [2]

A

B

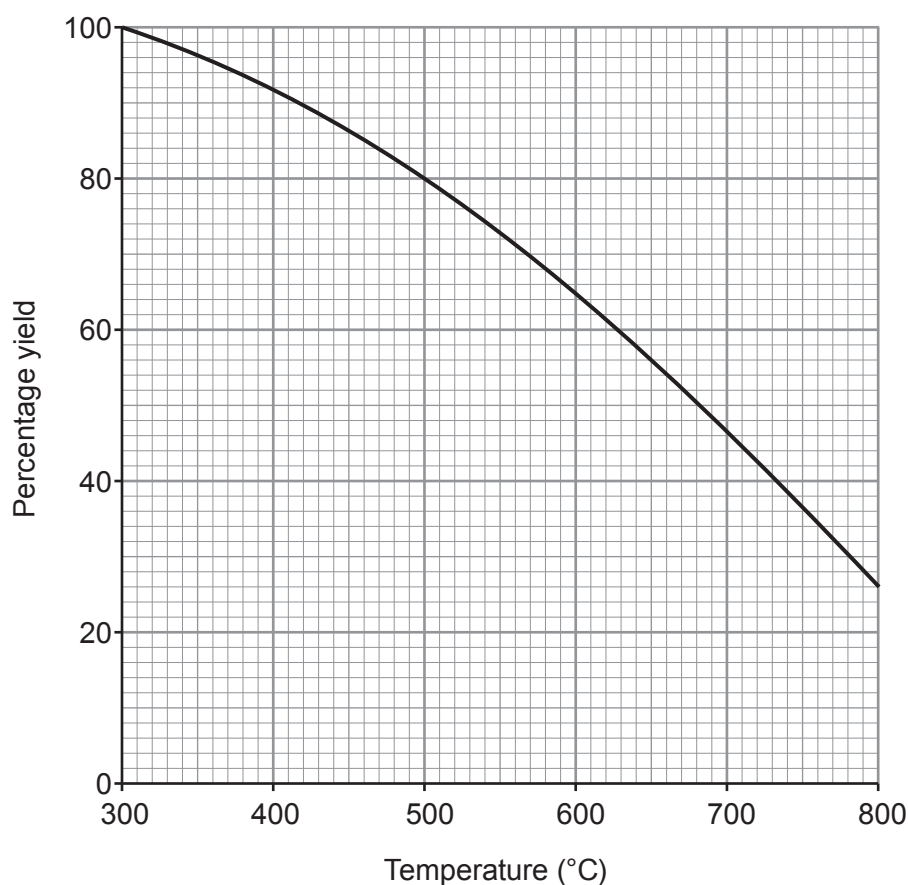
- (ii) State the purpose of vanadium(V) oxide in stage 2. [1]

.....

- (iii) Complete and balance the equation for the reaction in stage 2. [2]



- (iv) The graph shows how the percentage yield of sulfur trioxide changes with temperature between 300 °C and 800 °C.



Use the graph to answer parts I and II.

- I. State the trend in the percentage yield of sulfur trioxide as the temperature increases. [1]

.....

- II. Give the temperature **range** to be used to obtain a yield of sulfur trioxide greater than 80 %. [1]

..... to °C

- (v) One molecule of sulfur trioxide reacts with one molecule of sulfuric acid to form one molecule of oleum as the **only product**.

Complete the equation for this reaction by giving the formula of oleum. [1]



- (b) The photograph shows the exothermic reaction between concentrated sulfuric acid and sugar, $C_{12}H_{22}O_{11}$.



- (i) Name the **two** products formed. [2]

..... and

- (ii) State the type of reaction taking place. [1]

.....



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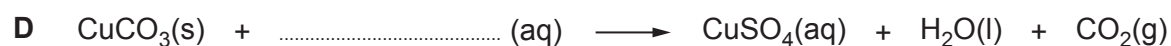
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3. (a) **A, B, C** and **D** are equations for reactions used to make salts.

(i) Complete the equation for each reaction.

[4]



(ii) Outline the **three** steps you would carry out to get a **pure** sample of silver chloride, AgCl, from the reaction mixture in equation **B**.

[2]

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- (b) Sodium chloride is made when sodium hydroxide reacts with hydrochloric acid.

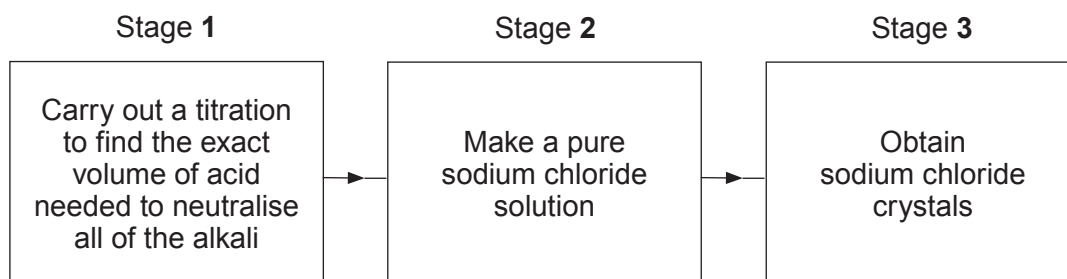


- (i) Write a balanced equation for the reaction.

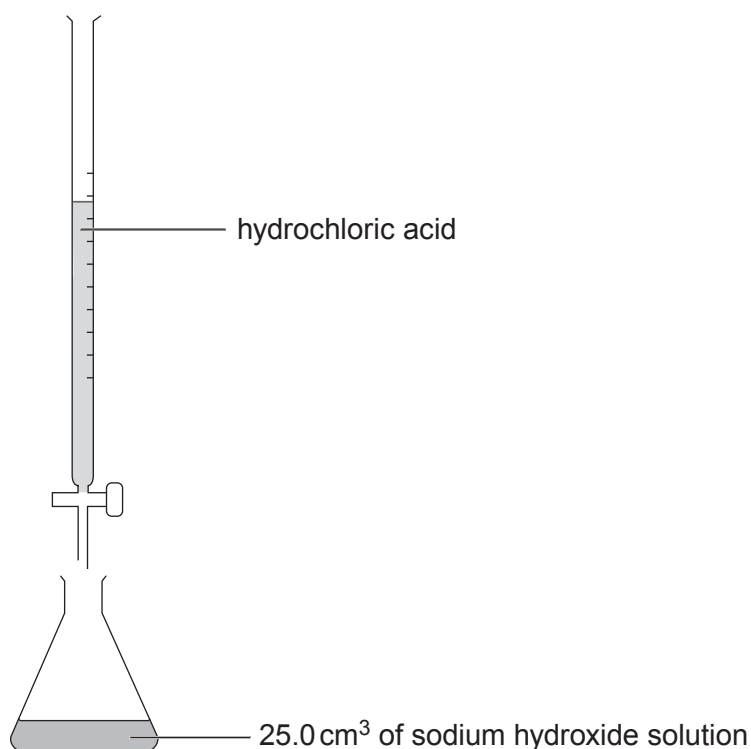
[2]

- (ii) A student was asked to prepare pure sodium chloride crystals using 25.0 cm³ of sodium hydroxide solution.

The flow diagram shows the student's plan.



- I. The student sets up the apparatus below to carry out stage 1.



Describe how the student will find the exact volume of acid needed to neutralise all of the alkali.

[2]

.....

.....

- II. The neutral solution obtained in stage 1 is not pure.

Describe what the student must do in stage 2 to make a pure sodium chloride solution.

[2]

.....

.....

.....

- III. State what the student must do in stage 3 to obtain sodium chloride crystals.

[1]

.....

.....

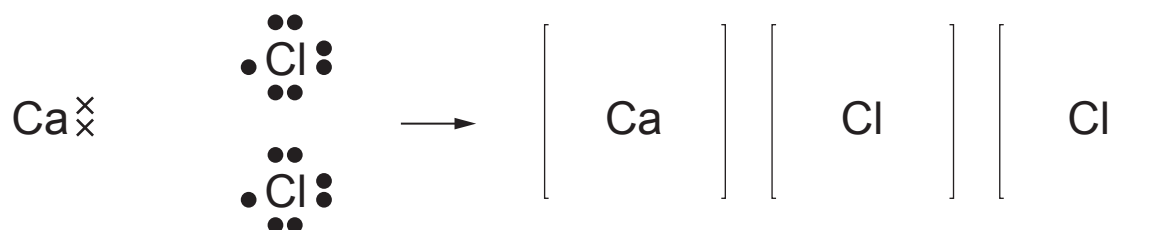


4. (a) Calcium reacts with chlorine to form the ionic compound calcium chloride.

Complete the dot and cross diagram to show the electronic changes that take place when calcium reacts with chlorine to form calcium chloride.

Show the charge on each ion.

[3]



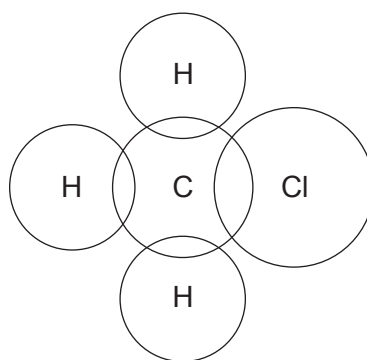
- (b) Complete the dot and cross diagram to show the bonding in a molecule of chloromethane.

[2]

hydrogen 1

carbon 2,4

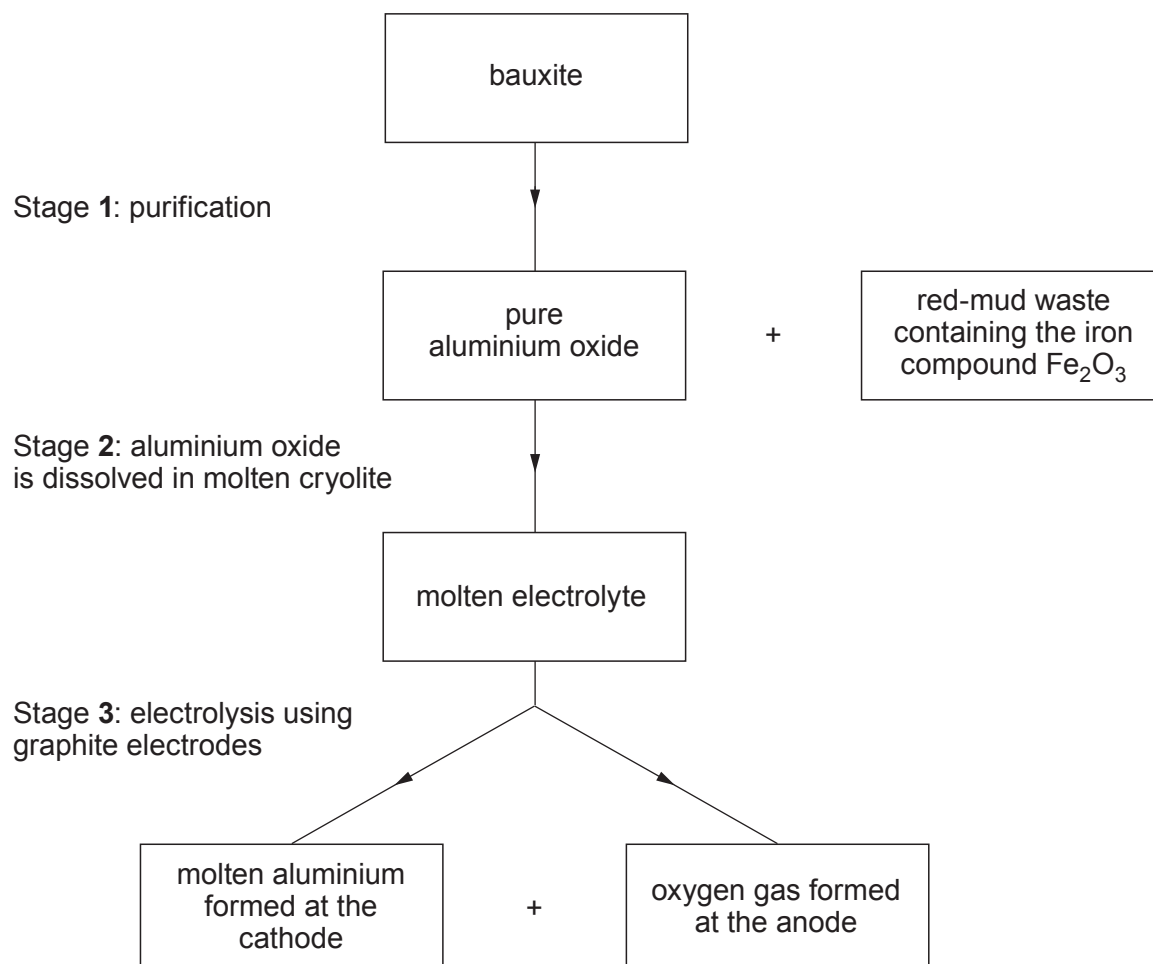
chlorine 2,8,7



[6 QER]



5. (a) The flow chart shows the stages involved in the extraction of aluminium from bauxite. Bauxite is a mixture of mainly aluminium oxide and a compound of iron.



- (i) Underline the chemical name for the iron compound, Fe_2O_3 , found in the red-mud waste. [1]

iron(V) oxide

iron(II) oxide

iron(III) oxide

iron(I) oxide

- (ii) Give **one** reason why aluminium oxide is dissolved in molten cryolite in stage 2. [1]

.....
.....

- (iii) Explain why the anode is regularly replaced in stage 3. [2]

.....
.....

- (iv) Complete and balance the equation for the electrolysis of aluminium oxide. [2]



- (b) The recycling rate for aluminium drinks cans in the UK reached 72 % in 2017.

Suggest why most aluminium waste is recycled. [2]

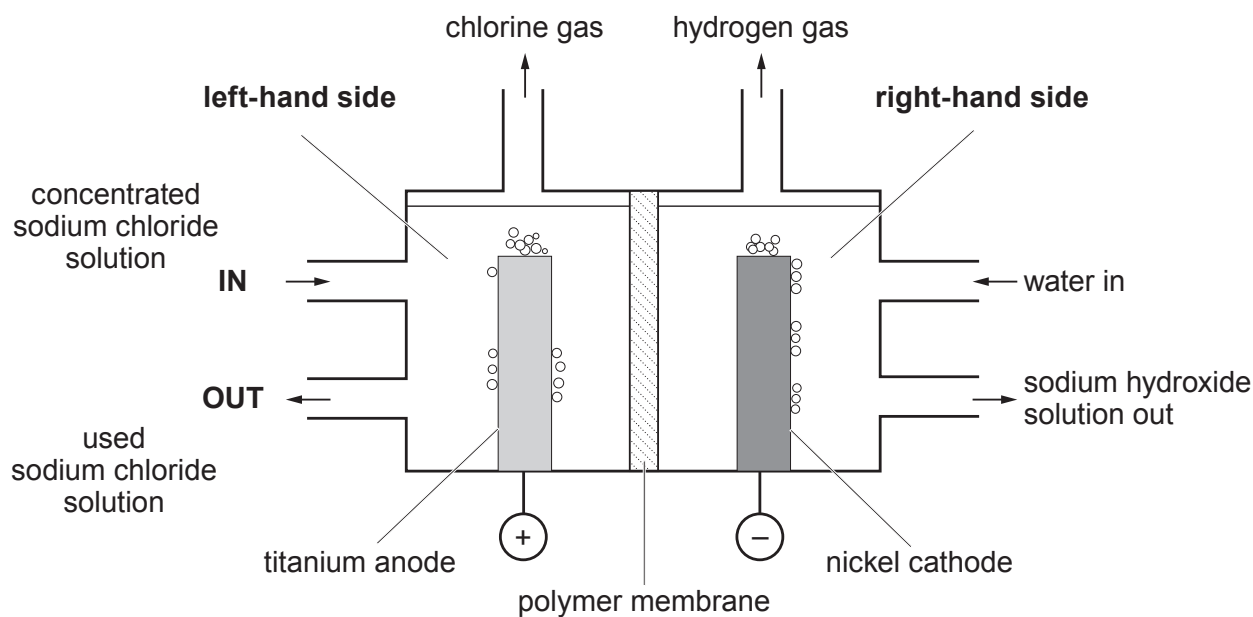
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- (c) The diagram shows a model of a membrane cell used to manufacture sodium hydroxide solution, chlorine and hydrogen by the electrolysis of aqueous sodium chloride.

The membrane cell is designed so that chlorine is kept separate from hydrogen and aqueous sodium hydroxide.

The polymer membrane only allows positive ions to pass through it. Sodium ions and hydrogen ions pass through from the left-hand side to the right-hand side of the cell.



Chlorine is an extremely corrosive substance which reacts with most metals. Titanium reacts quickly with chlorine gas but it resists attack by chlorine in aqueous conditions.



- (i) Put a **tick (✓)** in the box next to the correct list of ions found on either side of the membrane during the process. [1]

Left-hand side of the membrane	Right-hand side of the membrane
Na ⁺ Cl ⁻ H ⁺ OH ⁻	Cl ⁻ H ⁺ OH ⁻
H ⁺ OH ⁻ Cl ⁻	Na ⁺ H ⁺ OH ⁻
Na ⁺ Cl ⁻ H ⁺ OH ⁻	Na ⁺ H ⁺ OH ⁻
Na ⁺ Cl ⁻	Na ⁺ OH ⁻

☐☐☐☐

- (ii) Put a **tick (✓)** in the box next to the correct statement. [1]

titanium burns in chlorine in aqueous conditions

☐

titanium doesn't react with chlorine under any conditions

☐

aqueous conditions prevent chlorine from reacting with titanium

☐

aqueous conditions prevent chlorine from reacting with sodium chloride

☐

- (iii) Put a **tick (✓)** in the box next to the two substances formed on the right-hand side of the membrane if the sodium chloride solution were contaminated with magnesium bromide. [1]

sodium hydroxide and magnesium hydroxide

☐

sodium hydroxide and sodium bromide

☐

sodium hydroxide and magnesium bromide

☐

magnesium chloride and sodium

☐

6. (a) The table shows the formulae of the first four members of the alcohol family.

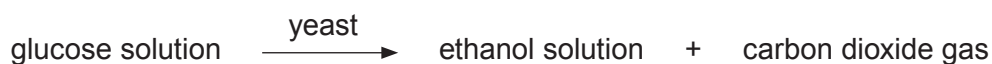
Alcohols
CH ₃ OH
C ₂ H ₅ OH
C ₃ H ₇ OH
C ₄ H ₉ OH

Give the general formula for the alcohol family.

[1]

.....

- (b) The following word equation represents the formation of ethanol by fermentation.



- (i) Complete and balance the equation for the production of ethanol and carbon dioxide from glucose.

Include the state symbols for the reactant and the products.

[3]



- (ii) Yeast contains a substance which catalyses the breakdown of glucose.

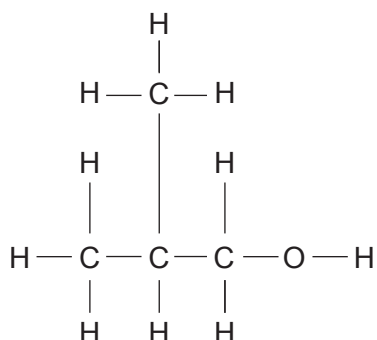
Give the **general** name of this type of catalyst.

[1]

.....



- (c) (i) C_4H_9OH has four isomers. The diagram below shows the structure of one of the isomers.



Put a **tick (✓)** in the box next to the name of this isomer.

[1]

butan-1-ol

☐

2-methylpropan-1-ol

☐

butan-2-ol

☐

2-methylpropan-2-ol

☐

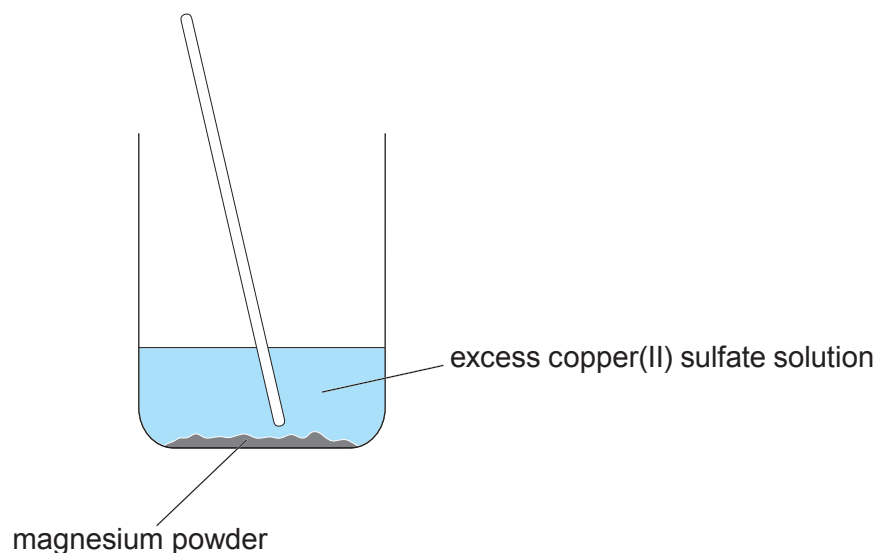
- (ii) Pentan-3-ol is one of the isomers of $C_5H_{11}OH$.

Draw the structure of pentan-3-ol.

[1]



7. (a) A student investigated the mass of copper formed when magnesium powder was added to excess copper(II) sulfate solution.



The student added four different masses of magnesium powder to separate 50 cm³ samples of copper(II) sulfate solution.

The table shows the results obtained.

Mass of magnesium added (g)	Mass of copper formed (g)
0.00	0.00
0.05	0.13
0.10	0.26
0.15	0.39
0.20	0.52

- (i) Give **one** observation that would show that copper(II) sulfate is always in excess in this investigation. [1]

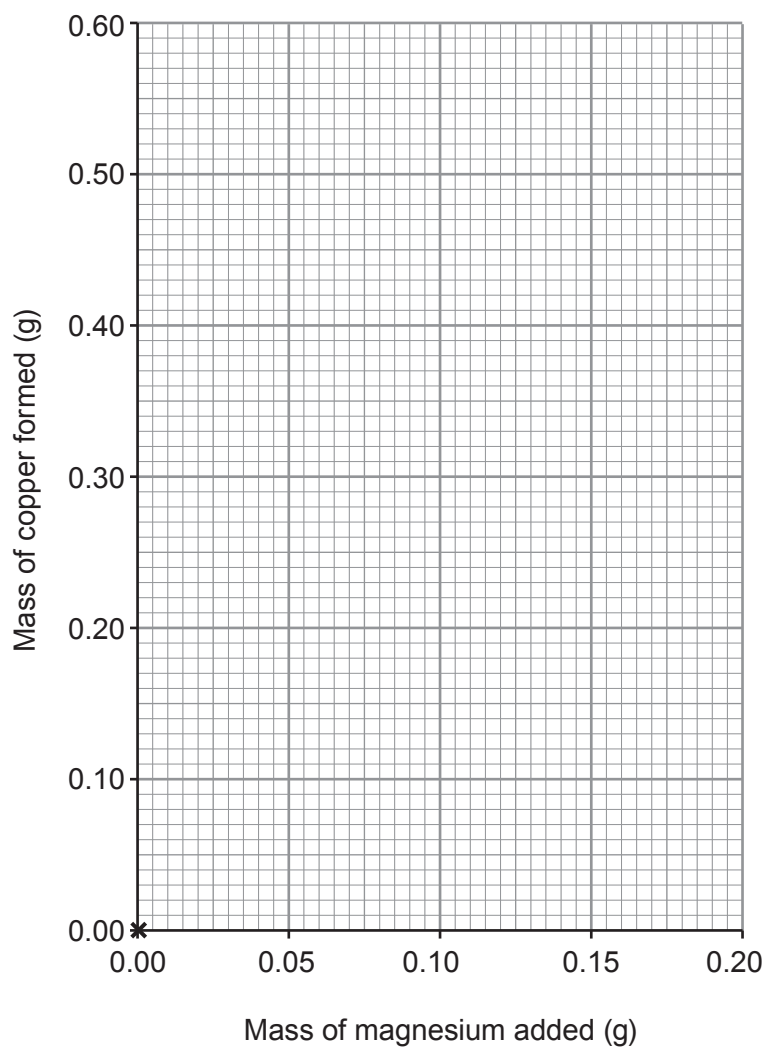
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- (ii) Plot the results from the table on the grid and draw a suitable line. [3]

The point (0,0) has been plotted for you.



- (iii) I. Use the graph to find the mass of magnesium needed to form 0.18 g of copper. [1]

..... g

- II. The student stated that 0.78 g of copper would be formed when 0.30 g of magnesium is used. Suggest how she arrived at this value. [1]

.....

.....

.....



- (b) To ensure that the copper(II) sulfate solution was always in excess the technician dissolved 3.19 g of CuSO_4 in 500 cm^3 of water.

- (i) Calculate the number of moles of copper(II) sulfate in 3.19 g. [2]

$$A_r(\text{O}) = 16 \qquad A_r(\text{S}) = 32 \qquad A_r(\text{Cu}) = 63.5$$

Number of moles = mol

- (ii) Use the equation below to calculate the concentration of the copper(II) sulfate solution in mol/dm^3 . [2]

$$\text{concentration (mol/dm}^3\text{)} = \frac{\text{number of moles}}{\text{volume (dm}^3\text{)}}$$

Concentration = mol/dm^3



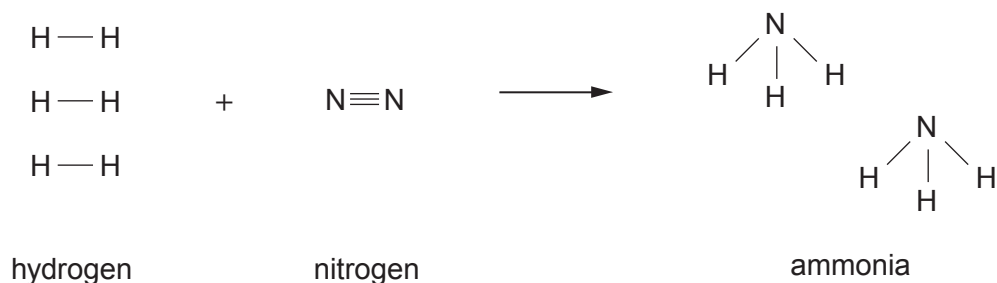
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8. Ammonia is manufactured from hydrogen and nitrogen in the Haber process.

- (a) The equation shows the bonds which are broken and the bonds which are formed in the production of ammonia.

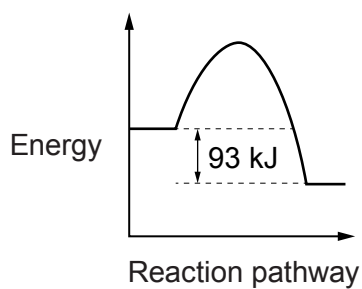


The bond energy of a H—H bond is 436 kJ. The energy needed to break **all** the bonds in the **reactants** is 2253 kJ.

- (i) Calculate the energy needed to break a N≡N bond. [2]

Energy = kJ

- (ii) The diagram below is the energy profile for the reaction.



Calculate the energy released when **all** the bonds in the ammonia molecules are formed. [1]

Energy = kJ



- (b) The table below shows the percentage yield of ammonia under different pressure and temperature conditions.

Pressure (atm)	Temperature (°C)			
	200	300	400	500
	Percentage yield of ammonia			
100	81.7	52.5	25.2	10.6
200	89.0	66.7		18.3
400	94.6	79.7	55.4	31.9
1000	98.3	92.6	79.8	57.5

- (i) Describe the relationship between the percentage yield of ammonia and the temperature.

[1]

- (ii) **Circle** the most likely percentage yield of ammonia if the process were carried out at 200 atm and 400 °C.

[1]

20 %

30 %

40 %

50 %



- (c) A student carries out a series of chemical tests on solutions of three unknown compounds, **A**, **B** and **C**. Her results are recorded in the table.

Compound	Flame test	Add NaOH(aq)	Add HCl(aq)	Add BaCl ₂ (aq)	Add AgNO ₃ (aq)
A	no colour	pungent gas given off turns damp red litmus paper blue	no reaction	no reaction	white precipitate forms
B	green flame	blue precipitate forms	no reaction	white precipitate forms	no reaction
C	lilac flame	no reaction	gas given off turns limewater milky	no reaction	no reaction

Use the information provided to identify compounds **A**, **B** and **C**.

[3]

A

B

C

END OF PAPER



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FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

0

1 H Hydrogen 1																																																																						
7	Li	3	9	Be	4	Beryllium																																																																
23	Na	11	24	Mg	12	Magnesium																																																																
39	K	19	40	Ca	20	Calcium	45	Sc	21	Scandium	48	Ti	22	Titanium	51	V	23	Vanadium	52	Cr	24	Chromium	55	Mn	25	Manganese	56	Fe	26	Iron	59	Co	27	Cobalt	59	Ni	28	Nickel	63.5	Cu	29	Copper	65	Zn	30	Zinc	70	Ga	31	Gallium	73	Ge	32	Germanium	75	As	33	Arsenic	79	Se	34	Selenium	80	Br	35	Bromine	84	Kr	36	Krypton
86	Rb	37	88	Sr	38	Strontium	89	Y	39	Yttrium	91	Zr	40	Zirconium	93	Nb	41	Niobium	96	Mo	42	Molybdenum	99	Tc		Technetium	101	Ru	44	Ruthenium	103	Rh	45	Rhodium	106	Pd	46	Palladium	108	Ag	47	Silver	112	Cd	48	Cadmium	115	In	49	Indium	119	Sn	50	Tin	122	Sb	51	Antimony	128	Te	52	Tellurium	127	I	53	Iodine	131	Xe	54	Xenon
133	Cs	55	137	Ba	56	Barium	139	La	57	Lanthanum	179	Hf	72	Hafnium	181	Ta	73	Tantalum	184	W	74	Tungsten	186	Re	75	Rhenium	190	Os	76	Osmium	192	Ir	77	Iridium	195	Pt	78	Platinum	197	Au	79	Gold	201	Hg	80	Mercury	204	Tl	81	Thallium	207	Pb	82	Lead	209	Bi	83	Bismuth	210	Po	84	Polonium	210	At	85	Astatine	222	Rn	86	Radon
223	Fr	87	226	Ra	88	Radium	227	Ac	89	Actinium																																																												

Key

relative atomic mass

atomic number